

Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Consider the 2D curve $\Gamma = \{(x, \sin x) : -\frac{\pi}{2} \leq x \leq \frac{7\pi}{2}\}$. A point mass m slides along Γ under the acceleration g of gravity and the normal force (oriented up) of Γ on m . Whenever the normal force vanishes, m deviates from Γ and comes solely under the influence of g , until re-contacting the ramp. Assume restitution is zero. The point mass starts at the left endpoint of Γ and is imparted v_0 horizontal speed to the right. We are interested only in scenarios where v_0 is calibrated precisely so that m launches off the first hump of Γ , clears the trough centered at $x = 3\pi/2$, and has its first contact-after-launch on the down-ramp of Γ on $(5\pi/2, 7\pi/2)$. Find v_0 that minimizes the magnitude of m 's incoming normal velocity at re-contact. Round your answer to 4 significant figures.

Solution — Let $x = a$ be the particle's launch point. We first work to find a a as a function of v_0 . While the particle is constrained to $y = \sin x$, conservation of energy determines the magnitude of its velocity at x by

$$v^2(x) = v_0^2 - 2g(\sin x + 1)$$

. By the standard unit normal formula for a parametrized curve, the upward unit normal at $(x, \sin x)$ is

$$\hat{\mathbf{n}} = \frac{(-\cos x, 1)}{\sqrt{1 + \cos^2 x}}$$

At any point x along the $y = \sin x$, the normal component of acceleration $a_n = v^2\kappa$ (where κ is 2D curvature) is

$$\frac{-v^2 \sin x}{(1 + \cos^2 x)^{3/2}}$$

(This is the normal acceleration required to follow the curve). The gravitational component of acceleration perpendicular to the ramp is

$$a_g = (0, -g) \cdot \hat{\mathbf{n}} = \frac{-g}{\sqrt{1 + \cos^2 x}}$$

The gravitational normal acceleration and the normal acceleration supplied by the ramp's normal force must sum to the net normal acceleration required to follow the curve:

$$\frac{N}{m} + \frac{-g}{\sqrt{1 + \cos^2 x}} = -\frac{v^2 \sin x}{(1 + \cos^2 x)^{3/2}}$$

. At launch $x = a$, the normal force vanishes ($N = 0$), giving:

$$v_a^2 = g \left(\frac{1 + \cos^2 x}{\sin x} \right)$$

where v_a is speed at loss of contact. From conservation of energy, we obtain

$$v_a^2 = v_0^2 - 2g(\sin x + 1)$$